Passive tools (theHarvester, whois, nslookup, Sublist3r

example.com, google.com or microsoft.com

Active tools (Nmap, Nikto, NetDiscover)

scanme.nmap.org for Nmap

1. theHarvester (finds emails, subdomains, hosts)

sudo apt update

sudo apt install theharvester -y

theHarvester --version

theHarvester --help

theHarvester -d microsoft.com -b crtsh

theHarvester -d microsoft.com -b anubis

theHarvester -d microsoft.com -b all

2. NetDiscover (finds live devices on your LAN)

ip a

sudo netdiscover -r 192.168.56.0/24

3. whois (domain registration info)

whois example.com

4. nslookup (DNS records)

nslookup example.com # IP address (A record)

nslookup 8.8.8.8 # reverse lookup

nslookup -type=MX gmail.com # mail servers

nslookup -type=NS microsoft.com # name servers

nslookup -type=TXT example.com # SPF and other text records

5. Sublist3r (finds subdomains)

sudo apt install sublist3r -y

sublist3r -d example.com

6. Nmap (finds open ports, services and OS)

nmap 192.168.56.10 # basic port scan(**VirtualBox Host-Only network**.)

nmap -sn 192.168.56.0/24 # find live hosts only

nmap -sV 192.168.56.10 # service versions

sudo nmap -O 192.168.56.10 # OS detection

sudo nmap -A 192.168.56.10 # all of the above

1. Is the host alive? (no port scan)

nmap -sn scanme.nmap.org

2. Basic scan (top 1000 ports)

nmap scanme.nmap.org

3. Scan specific ports only

nmap -p 22,80,443 scanme.nmap.org

4. Service versions

nmap -sV scanme.nmap.org

5. OS detection (needs sudo)

sudo nmap -O scanme.nmap.org

6. Aggressive scan = version + OS + scripts + traceroute

sudo nmap -A scanme.nmap.org

8.8.8.8

8. Wireshark (captures packets)

sudo wireshark &

1. Double-click eth0 to start capturing.
2. Open a site or run ping to create traffic.
3. Click the red square to stop.

ping -c 4 8.8.8.8

Useful filters: http, dns, icmp, tcp

9. Armitage (GUI for Metasploit)

sudo systemctl start postgresql

sudo msfdb init

sudo armitage

Click **Connect** → **Yes**. Then **Hosts → Nmap Scan → Quick Scan** → enter the lab IP → **Attacks → Find Attacks**.
Inference: it shows the live hosts, their OS and services, and the matching exploits. This is why unpatched services are risky.

3. Maltego (Exp 3a)

What it does: Gathers OSINT and shows it as a **visual graph** of relationships (domain → subdomains → IPs → emails).

Steps:

1. Install **Maltego CE** (free), register on the Maltego site, and log in.
2. **New graph** (the blank-page icon).
3. In the left **Entity Palette → Infrastructure**, drag **Domain** onto the graph.
4. Double-click it and change the name to example.com.
5. **Right-click the domain → Run Transform** and pick from the list, for example:
    - **To DNS Name - NS** (name servers)
    - **To DNS Name - MX** (mail servers)
    - **To IP Address [DNS]**
    - **To Website [Quick lookup]**
6. Or right-click → **Run Machine → Footprint L1** to run several transforms automatically.

google.com

1. **Google Dorking**
2. **Uses special search operators to find public information about a domain. It's passive, so nothing touches the target directly.**

1. Find indexed pages

Shows pages Google has indexed from the domain.

site:google.com

2. Find PDF documents

site:google.com filetype:pdf

3. Find login-related pages

site:google.com inurl:login

4. Search page titles

site:google.com intitle:about

2. Shodan (Exp 3b)

Where: Browser, at **shodan.io**.
What it does: A search engine for internet-connected devices. It shows which hosts, open ports, services and software versions are publicly exposed. It has already scanned the target, so you send nothing to it.

Look up a domain or IP: type example.com or 8.8.8.8

3. Robtex

Where: Browser, at **robtex.com**.
What it does: DNS and network lookups. Enter a domain or IP and it shows how it's connected.

Type example.com (or an IP like 8.8.8.8)

2. FOCA

FOCA finds public files of a domain and reads the hidden **metadata** inside them.

Step 1: Create a project

1. Open FOCA → **Project → New project**.
2. Project name: anything (e.g. test).
3. Domain website: example.com (or the authorized domain given).

4. Choose a folder to save to → **Create**.

Step 2: Search for documents

- Tick the search engines (Google, Bing) and the extensions (doc, docx, pdf, xls, xlsx, ppt and so on).
- Click **Search All**.
- Files appear in the list with ID, Type, URL and Size.

Step 3: Download and extract (from your slide)

1. Select the files in the list.
2. Right-click → **Download All**.
3. When the Download column shows ✓, right-click → **Extract All Metadata**.

site:google.com filetype:pdf

3. In FOCA, right-click in the file list → **Add folder** (or **Add file**). This option is in the right-click menu in your slide.
4. Select the files → right-click → **Extract All Metadata**.
5. Open **Metadata Summary** → **Users, Software, Operating Systems, Folders**.

nikto -h http://127.0.0.1

Cybersecurity Tools Practical Guide

Procedure: 4 Steps for Each Tool + Expected Output

Below are the procedures for all **15 tools**, written in simple steps suitable for your practical exam.

For each tool, you will get:

1. Four procedure steps.
2. The command or action to perform.
3. An example of the output you should expect.

4. A short inference you can write in your record.

Important: Example outputs below illustrate realistic output formats. Actual results depend on your network, software version, permissions, search sources, and the target's current configuration. I won't present invented scan results as if they were actual results from your computer.

Before starting

Your commands use Linux syntax, so open a Linux terminal, such as Kali Linux or Ubuntu.

- Use example.com, google.com, or microsoft.com for appropriate passive lookups.
- Use scanme.nmap.org for Nmap practice within its published scanning policy.
- Use your own lab IP addresses, such as 192.168.56.10, for active network scans.
- Only scan systems for which you have authorization.

---

1. TheHarvester

Aim: Collect publicly available email addresses, hostnames, and subdomains associated with a domain.

Procedure (4 steps)

Step 1: Install the tool.

sudo apt update

sudo apt install theharvester -y

Step 2: Check the version and available options.

theHarvester --version

theHarvester --help

Step 3: Search a domain using a supported source.

theHarvester -d microsoft.com -b crtsh

Step 4: Examine the results.

Look for discovered email addresses, hostnames, and other information returned by the selected source.

Expected output

The output format may look like this:

[*] Target: microsoft.com

[*] Hosts found:

example.microsoft.com

mail.microsoft.com

[*] Emails found:

user@microsoft.com

These entries illustrate the output format; they are not verified findings.

The actual search may return different results or none. If the source is unsupported, unavailable, or requires configuration, the command may fail.

Inference: TheHarvester gathers publicly available information that can help map an organization's online presence.

---

2. Netdiscover

Aim: Discover devices on a local network using ARP.

Procedure (4 steps)

Step 1: Open a terminal and inspect network interfaces.

ip a

Identify the interface and subnet used by your lab network.

Step 2: Determine your lab subnet.

For example, if your lab network is 192.168.56.0/24, use that subnet only if it matches your actual network.

Step 3: Run Netdiscover against the lab subnet.

sudo netdiscover -r 192.168.56.0/24

Step 4: Examine the discovered devices.

Look at the IP address, MAC address, and vendor fields.

Expected output

IP At MAC Address Count Len MAC Vendor

192.168.56.1 AA:BB:CC:DD:EE:01 1 60 Example Vendor

192.168.56.10 AA:BB:CC:DD:EE:02 1 60 Example Vendor

The addresses and vendor names are illustrative.

If no devices appear, check that the interface is connected to the correct network and that the subnet is correct.

Inference: Netdiscover identifies devices responding to ARP requests on the local network.

---

3. WHOIS

Aim: Obtain domain registration information.

Procedure (4 steps)

Step 1: Open a Linux terminal.

Step 2: Query a domain.

whois example.com

Step 3: Examine the returned registration details.

Look for the registrar, dates, domain status, and nameservers.

Step 4: Record the available information.

Some registrant details may be hidden by privacy protection.

Expected output

A response may contain fields similar to these:

Domain Name: EXAMPLE.COM

Registrar: Example Registrar

Creation Date: YYYY-MM-DD

Registry Expiry Date: YYYY-MM-DD

Name Server: NS1.EXAMPLE.NET

Domain Status: clientTransferProhibited

The registrar and dates above are placeholders, not actual registration details.

Inference: WHOIS provides domain registration information that helps investigate a domain's administrative details.

---

4. Nslookup

Aim: Query DNS records and resolve domain names.

Procedure (4 steps)

Step 1: Find the IP address of a domain.

nslookup example.com

Step 2: Perform a reverse lookup for an IP address.

nslookup 8.8.8.8

Step 3: Query the mail servers of a domain.

nslookup -type=MX gmail.com

Step 4: Query nameserver and TXT records.

nslookup -type=NS microsoft.com

nslookup -type=TXT example.com

Expected output

For the first command, you may see a response similar to:

Server: 192.168.1.1

Address: 192.168.1.1#53

Non-authoritative answer:

Name: example.com

Address: 93.184.216.34

The IP address shown is an illustrative example and may not reflect the current DNS answer.

Other commands return record types such as:

- **PTR:** A hostname associated with an IP address, if available.
- **MX:** Mail servers.
- **NS:** Authoritative nameservers.
- **TXT:** Text records, which may include SPF information.

Inference: Nslookup retrieves DNS information used to resolve domain names and investigate DNS configuration.

---

5. Sublist3r

Aim: Discover subdomains associated with a domain.

Procedure (4 steps)

Step 1: Install Sublist3r.

sudo apt update

sudo apt install sublist3r -y

If the package is unavailable in your Linux distribution, use its official project installation instructions.

Step 2: Check the available options.

sublist3r -h

Step 3: Search a domain you are authorized to investigate.

sublist3r -d example.com

Step 4: Examine the discovered subdomains.

Record the subdomains displayed by the tool.

Expected output

The output may look like this:

[-] Enumerating subdomains now for example.com

www.example.com

mail.example.com

blog.example.com

[-] Total Unique Subdomains Found: 3

The subdomains and count are illustrative, not verified results. A real run may return different results or zero subdomains.

Inference: Sublist3r enumerates subdomains that may help build an inventory of an organization's Internet-facing assets.

---

6. Nmap

Aim: Discover live hosts, open ports, services, and possible operating systems.

Procedure (4 steps)

Step 1: Check whether the host is reachable.

Use the authorized Nmap test host:

nmap -sn scanme.nmap.org

This checks host discovery without performing a normal port scan.

Step 2: Perform a basic scan.

nmap scanme.nmap.org

This scans the default set of commonly scanned TCP ports.

Step 3: Detect service versions.

nmap -sV scanme.nmap.org

Step 4: Scan specific ports or perform OS detection on an authorized lab host.

For example:

nmap -p 22,80,443 scanme.nmap.org

For a lab machine where you have permission:

sudo nmap -O 192.168.56.10

Expected output

A scan may display a table like this:

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

443/tcp closed https

Important: This table is illustrative, not a report of the current state of scanme.nmap.org.

The actual output may contain different ports and states:

- open: A service is accepting connections.
- closed: The host is reachable, but no service is listening on that port.
- filtered: Filtering prevents Nmap from determining whether the port is open.

OS detection may report a probable operating system or say that the result is inconclusive.

Inference: Nmap identifies hosts, accessible ports, and services to support network inventory and security assessment.

---

7. Wireshark

Aim: Capture and analyze network packets.

Procedure (4 steps)

Step 1: Open Wireshark.

sudo wireshark

If it is not installed:

sudo apt update

sudo apt install wireshark -y

Step 2: Select the correct network interface.

Double-click the interface carrying your traffic, such as eth0 or wlan0.

Step 3: Generate network traffic.

Open a website or run:

ping -c 4 8.8.8.8

Step 4: Stop the capture and apply a display filter.

Click the red stop button and try these filters individually:

dns

icmp

tcp

Expected output

Wireshark displays captured packets in a table.

No.	Source	Destination	Protocol	Info
1	Your device	DNS server	DNS	Standard query
2	DNS server	Your device	DNS	Standard query response
3	Your device	8.8.8.8	ICMP	Echo request
4	8.8.8.8	Your device	ICMP	Echo reply

These rows illustrate the output format. Your actual capture will depend on the traffic observed.

Inference: Wireshark captures packets and displays their protocol information, allowing network communication to be examined.

---

8. Armitage

Aim: Visualize hosts and manage authorized penetration-testing activities using Metasploit.

Procedure (4 steps)

Step 1: Start PostgreSQL.

On a Kali Linux installation with PostgreSQL configured:

sudo systemctl start postgresql

Step 2: Initialize the Metasploit database if necessary.

sudo msfdb init

If the database is already initialized, you may not need to repeat this step.

Step 3: Launch Armitage.

sudo armitage

Click **Connect** when prompted. If asked whether to start Metasploit, follow the prompt.

Step 4: Scan an authorized lab target.

In Armitage, use the host scanning menu, such as **Hosts → Nmap Scan → Quick Scan**, if available in your version. Enter the IP address of your lab machine.

Review the discovered hosts and services. Do not launch attacks against systems without explicit authorization.

Expected output

Armitage may display:

- A graphical representation of discovered hosts.
- Host IP addresses.
- Detected services and ports.
- OS information when detected.
- Available sessions, if a test has successfully created one.

Inference: Armitage presents network discovery and penetration-testing information in a graphical interface. Finding a possible exploit does not prove that the target is vulnerable.

---

9. Maltego

Aim: Gather OSINT and visualize relationships between domains, DNS records, IP addresses, and other entities.

Procedure (4 steps)

Step 1: Open Maltego.

Install the available Maltego Community Edition or other appropriate edition from its official website. Register and sign in if required.

Step 2: Create a graph.

Open a new graph and locate **Domain** in the Entity Palette.

Step 3: Add a domain.

Drag the Domain entity onto the graph and enter a domain such as:

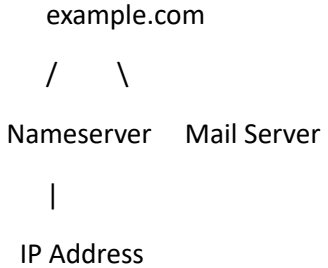
example.com

Step 4: Run available transforms.

Right-click the domain and select an appropriate transform, such as one that retrieves DNS nameservers, mail servers, IP addresses, or website information. Transform names and availability depend on the Maltego version and connected data sources.

Expected output

Maltego creates a visual graph that might look like this:



The diagram illustrates possible relationships, not actual findings for example.com.

Inference: Maltego connects information from supported sources and presents relationships in a visual graph.

---

10. Google Dorking

Aim: Find specific publicly indexed information using advanced Google search operators.

Procedure (4 steps)

Step 1: Open Google in your browser.

Step 2: Search for indexed pages belonging to a domain.

site:example.com

Step 3: Search for documents or particular pages.

For PDFs:

site:example.com filetype:pdf

For pages with login-related URLs:

site:example.com inurl:login

Step 4: Examine the search results.

Open relevant public pages and record their titles and URLs. Do not attempt to access restricted resources or use exposed credentials.

Expected output

Google may return results in a format similar to:

Example Domain

https://example.com/

Example Document

https://example.com/sample.pdf

These URLs illustrate the format of search results; they are not claims that Google currently returns those particular results for every query.

Inference: Google Dorking uses search operators to locate information that a search engine has indexed.

---

11. Shodan

Aim: Investigate publicly exposed Internet-connected devices and services.

Procedure (4 steps)

Step 1: Open the official website.

Visit Shodan in your browser.

Step 2: Sign in if required.

Some searches and features may require an account or paid access.

Step 3: Search for a domain or IP address.

For example, enter:

example.com

You can also search for an IP address you are authorized to investigate.

Step 4: Review the available host information.

Examine reported ports, services, banners, hostnames, and observation dates.

Expected output

A host record may contain fields similar to:

IP: 203.0.113.10

Port: 443

Service: HTTPS

Hostname: server.example.com

These values are illustrative, not a verified Shodan result.

If no records appear, Shodan may not have collected information for that host, or the information may not be available to your account.

Inference: Shodan searches previously collected Internet-facing device information; its results do not guarantee that a service is still accessible.

---

12. Robtex

Aim: Investigate DNS information and relationships between domains and IP addresses.

Procedure (4 steps)

Step 1: Open the official website.

Visit Robtex in your browser.

Step 2: Enter a domain or IP address.

For example:

example.com

Step 3: Review the available information.

Look for DNS records, IP addresses, nameservers, mail servers, and related infrastructure.

Step 4: Record the relevant relationships.

Note the domain, associated IP addresses, and DNS information actually displayed.

Expected output

The results may contain information similar to:

Domain: example.com

IP Address: [returned IP address]

Nameservers: [returned nameservers]

Mail Servers: [returned mail servers]

The bracketed fields are placeholders for the actual returned values.

Inference: Robtex helps investigate relationships between Internet domains, IP addresses, and DNS infrastructure.

---

13. FOCA

Aim: Find publicly available documents and extract metadata from supported files.

Your earlier screenshot showed FOCA 3.4.7.1 returning zero search results. Its older search integrations may not work reliably, so the local-file method is a useful alternative if supported by your installation.

Procedure (4 steps)

Step 1: Create a project.

Open FOCA and select **Project → New project**, if that option is available.

Enter:

- Project name: FOCA_Practice
- Domain website: a domain you are authorized to investigate
- Folder: a suitable location for downloaded documents

Click **Create**.

Step 2: Search for documents.

Select the available search engines and file extensions, such as PDF and DOCX. Click **Search All** or the equivalent search button.

Step 3: Download or import documents.

If results appear, select the documents and use the available download option. If the search returns zero results, try the local-file option supported by your version to add a document you own.

Step 4: Extract and inspect metadata.

Select the supported documents and use **Extract All Metadata** or the corresponding option. Open **Metadata Summary** if available.

Expected output

FOCA may display metadata fields similar to:

File: Report.docx

Author: Example User

Software: Microsoft Office

Creation Date: [recorded date]

The author and date above are illustrative. The actual metadata depends on the document; some files contain little or no useful metadata.

Inference: FOCA extracts metadata from documents, which can reveal information about how a document was created.

---

14. Nikto

Aim: Scan a web server for potentially dangerous files, outdated software, and common web server misconfigurations.

Procedure (4 steps)

Step 1: Install Nikto.

sudo apt update

sudo apt install nikto -y

Step 2: Check the available options.

nikto -Help

Step 3: Scan a local test web server.

If a web server is running on your own machine, use:

nikto -h http://127.0.0.1

Here, 127.0.0.1 refers to your own computer. The command will only produce useful results if a web server is running there.

Step 4: Review the findings.

Look for potentially risky files, server configuration issues, outdated components, and other reported findings.

Expected output

Nikto may display results in a format similar to:

- Nikto v2.x.x

- Target IP: 127.0.0.1

- Target Port: 80

- Server: Example Web Server

+ /: The X-Content-Type-Options header is not set.

+ /: The server may disclose version information.

These findings are illustrative, not actual results from your computer. A real scan may report different issues or no findings.

Inference: Nikto checks web servers for common security problems. It is more specifically focused on web servers than Nmap.

---

15. Sysmon

Aim: Monitor Windows system activities and record events useful for security investigations.

Note: Sysmon is a Windows tool, so you perform these steps on a Windows computer rather than in a Kali Linux terminal.

Procedure (4 steps)

Step 1: Download Sysmon.

Download it from the official Microsoft Sysinternals Sysmon page.

Extract the downloaded files.

Step 2: Install Sysmon.

Open PowerShell as Administrator, navigate to the extracted folder, and run:

.\Sysmon64.exe -i

If your installation already has Sysmon running, you may need to check its current configuration rather than reinstall it.

Step 3: Generate a system event.

Open Notepad or another ordinary application. Sysmon can record the process creation event if the relevant event is enabled.

Step 4: View the event logs.

1. Press Windows + R.
2. Enter eventvwr.msc.
3. Navigate to **Applications and Services Logs → Microsoft → Windows → Sysmon → Operational**.
4. Examine the available events.

Expected output

A process creation event may contain fields such as:

Event ID: 1

Image: C:\Windows\System32\notepad.exe

Process ID: 2450

Command Line: notepad.exe

These values illustrate the format. Your actual process ID and recorded details will differ.

Inference: Sysmon records selected Windows system activities, helping investigators understand which processes ran and what other monitored events occurred.

Final Quick Revision Table

Tool	Procedure performed	Output obtained
TheHarvester	Search a domain using a supported source	Emails, hosts, and related information
Netdiscover	Scan the local lab subnet using ARP	IP addresses and MAC addresses
WHOIS	Query a domain	Registration details
Nslookup	Query DNS records	IP addresses and DNS responses
Sublist3r	Enumerate a domain	Subdomain list
Nmap	Scan an authorized host	Host status, ports, services, and possible OS
Wireshark	Capture network traffic	Packet details
Armitage	Scan a lab host through its interface	Graphical host and service information
Maltego	Run transforms on a domain entity	Relationship graph
Google Dorking	Search with advanced operators	Indexed pages and documents
Shodan	Search a domain or IP	Previously collected device and service information
Robtex	Look up a domain or IP	DNS and infrastructure relationships
FOCA	Search/import documents and extract metadata	Document properties and metadata
Nikto	Scan a local test web server	Potential web server security issues

Tool	Procedure performed	Output obtained
Sysmon	Monitor Windows events	System event logs

Important for your practical exam: If your instructor asks you to explain the *procedure*, describe the actions you perform in sequence. If asked for the *output*, show the relevant fields or results produced by the tool. The sample outputs here demonstrate the expected format; your practical record should contain the actual output from your own run.